```
# Banking enterprise: maximum security, uses ML-KEM-1024 +
ML-DSA-87
engine = PQCEngine(DomainProfile.ENTERPRISE)

# Encrypt a payload
ciphertext, encapsulated_key = await engine.
encrypt(plaintext)

# Decrypt
recovered = await engine.decrypt(ciphertext, encapsulated_
key)
```

The engine supports hybrid classical/post-quantum key exchange — combining X25519 or P-384 ECDH with ML-KEM via hybrid schemes (X25519MLKEM768, P384MLKEM1024) — so existing TLS 1.3 stacks remain compatible while gaining quantum resistance.

NIST-standardised algorithms are:

- *FIPS 203:* ML-KEM (Kyber) 512, 768, and 1024 — key encapsulation for session keys
- *FIPS 204:* ML-DSA (Dilithium) at levels 2, 3, and 5 — digital signatures
- *FIPS 205:* SLH-DSA (SPHINCS+) — stateless hash-based signatures

Additional algorithms are:

- Falcon 512 and 1024 — compact signatures for bandwidth-constrained channels
- LMS (NIST SP 800-208, RFC 8554) and XMSS (RFC 8391) — stateful hash-based signatures

Advanced primitives are:

- Zero-knowledge proofs (ZKP) for privacy-preserving verification
- Verifiable random functions (VRF) and threshold signatures (t-of-n distributed signing)
- CNSA 2.0 support for defence and government deployments
- Crypto agility negotiation for seamless algorithm migration
- Encryption overhead on the hot path is under 1 millisecond.

Autonomous agents that respond in under a second

QBITEL Bridge's security response layer is built around a multi-agent architecture with 16+ specialised agents. Agents are workers with typed capabilities including threat analysis, protocol analysis, anomaly detection, incident response, and compliance auditing.

Each agent inherits from BaseAgent, which provides:

- Lifecycle management (start, stop, pause, resume, health check)
- Inter-agent communication via a typed message bus with cross-system routing
- Persistent memory with configurable retention policies
- Automatic Prometheus metrics instrumentation
- Circuit breaker pattern for fault tolerance

The *UnifiedAgentInterface* bridges three agent subsystems — core agents, legacy whisperer agents, and BPO-specific agents — through a common protocol with a central registry.

The LLM integration layer connects agents to local or cloud language models — including Ollama for fully air-gapped deployments. This enables agents to generate human-readable incident reports, suggest remediation steps, and

Figure 2: PQC algorithm selection by domain and constraint